

PROJECT REPORT

(Project Term June-July 2026)

E-commerce Website Automation

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DECLARATION

I hereby declare that the project work entitled (“Title of the project”) is an authentic record of my own work carried out as requirements of Project for the award of B.Tech Degree in _____(Programme Name) from Lovely Professional University, Phagwara, under the guidance of (Name of Faculty Mentor), during January to May 2025. All the information furnished in this project report is based on my own intensive work and is genuine.

Name of Student :Abhishek Kumar

Registration Number: 12404519

(Signature of Student)

Date:

CERTIFICATE

This is to certify that the declaration statement made by the student is correct to the best of my knowledge and belief. He has completed this Project under my guidance and supervision. The present work is the result of his original investigation, effort and study. No part of the work has ever been submitted for any other degree at any University. The Project is fit for the submission and partial fulfillment of the conditions for the award of B.Tech Degree in DevOps from Lovely Professional University, Phagwara.

Signature and Name of the Mentor

Designation

School of Computer Science and Engineering,
Lovely Professional University,
Phagwara, Punjab.

Date :

ACKNOWLEDGEMENT

I would like to express my heartfelt gratitude to everyone who contributed to the successful completion of this project.

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ABSTRACT

LoveParcel is a full-stack e-commerce web application designed for gifting and custom hamper experiences. This project focuses on applying **DevOps principles and practices** to automate the building, testing, and deployment of both the frontend and backend services of the platform.

The application is built using **Next.js 16** for the frontend and **Node.js (Express.js)** with **Prisma ORM** and **MongoDB Atlas** for the backend. The DevOps automation pipeline leverages **Docker** for containerization, **GitHub Actions** for Continuous Integration and Continuous Deployment (CI/CD), **Amazon ECR** for container image registry, and **AWS EC2** and **Kubernetes (EKS)** for cloud deployment.

Key features include product listing and search, custom hamper builder, cart management, multi-gateway payment integration (Razorpay and Cashfree), order tracking, Google OAuth authentication, coupon management, and cloud image storage via Cloudinary. The platform uses **Redis** (Upstash) for caching and session management, and **Nginx** as a reverse proxy with SSL termination.

The primary DevOps contribution of this project is the establishment of a fully automated CI/CD pipeline that enables zero-downtime deployments, automatic health checks, rollback capabilities, and environment-specific configuration management using Kubernetes Secrets.

CHAPTER 1 — INTRODUCTION

1.1 Project Overview

LoveParcel is a comprehensive gifting and e-commerce platform that allows users to browse curated gift products, build custom hampers, place orders, and make payments through multiple payment gateways. The platform caters to individual customers looking for personalized gifting experiences for various occasions such as birthdays, anniversaries, festivals, and corporate gifting.

From a **DevOps perspective**, this project demonstrates the end-to-end automation of the software delivery pipeline for a production-grade e-commerce application. The project covers:

- **Containerization** of both the frontend (Next.js) and backend (Express.js) applications using Docker
- **Multi-stage Docker builds** to produce optimized, minimal production images
- **Orchestration** using Docker Compose for local development and Kubernetes (AWS EKS) for production
- **CI/CD Pipeline** using GitHub Actions for automated testing, building, and deployment
- **Cloud Infrastructure** on Amazon Web Services (AWS) using EC2 and/or EKS
- **Reverse Proxy and Load Balancing** using Nginx
- **Secret Management** using Kubernetes Secrets and GitHub Actions Secrets
- **Auto-scaling** using Kubernetes Horizontal Pod Autoscaler (HPA)

1.2 Motivation

The rapid growth of the e-commerce industry in India has created a demand for reliable, scalable, and continuously available web applications. Traditional deployment methods — manual FTP uploads, SSH-based updates, or monolithic server setups — are no longer adequate for modern applications that require:

- **Zero-downtime deployments** during updates
- **Horizontal scalability** to handle traffic spikes (festivals, sales events)
- **Automated testing** to catch bugs before production
- **Fast rollbacks** when a deployment fails
- **Reproducible environments** to eliminate “works on my machine” problems

This project addresses all of the above by applying DevOps engineering principles to a real-world gifting e-commerce platform.

1.3 Objectives

The primary objectives of this project are:

1. **Develop** a fully functional e-commerce web application for gifting (LoveParcel)
2. **Containerize** the frontend and backend applications using Docker with multistage builds
3. **Implement CI/CD** using GitHub Actions for automated build, test, and deploy workflows
4. **Deploy** the application on AWS infrastructure (EC2 / EKS) with proper networking
5. **Configure** Nginx as a reverse proxy with HTTPS/SSL using Let's Encrypt
6. **Implement auto-scaling** using Kubernetes HPA based on CPU and memory metrics
7. **Manage secrets** securely using GitHub Actions Secrets and Kubernetes Secrets

8. **Enable** automated rollback on failed deployments

1.4 Technology Stack

Layer	Technology	Purpose
Frontend	Next.js 16, TypeScript, Tailwind CSS	User Interface
Backend	Node.js, Express.js	REST API Server
ORM	Prisma ORM	Database schema and queries
Database	MongoDB Atlas	Primary data store
Caching	Redis (Upstash)	Session and data caching
Auth	JWT, Google OAuth 2.0	Authentication
Payments	Razorpay, Cashfree	Payment processing
Media	Cloudinary	Image storage and CDN
Email	Nodemailer (SMTP)	Transactional emails
Containerization	Docker, Docker Compose	Application packaging
Orchestration	Kubernetes (AWS EKS)	Production container management
CI/CD	GitHub Actions	Automated pipelines
Registry	Amazon ECR	Docker image storage
Cloud	AWS EC2 / EKS	Cloud hosting
Reverse Proxy	Nginx	Traffic routing, SSL
SSL	Let's Encrypt (Certbot)	HTTPS certificates
Version Control	Git, GitHub	Source code management
State Management	Redux Toolkit	Frontend state

CHAPTER 2 — SCOPE OF THE STUDY (PROBLEM STATEMENT)

2.1 Problem Statement

In the modern digital economy, e-commerce businesses face several critical operational challenges:

1. Manual and Error-Prone Deployments Traditional deployment processes involve manually connecting to servers via SSH, copying files, installing dependencies, and restarting services. This process is: - Slow (30–60 minutes per deployment)

- Error-prone (human mistakes during configuration) - Inconsistent across environments (development, staging, production)

2. Environment Inconsistency Developers frequently encounter “works on my machine” problems where code behaves differently in development versus production. This is caused by differences in Node.js versions, operating systems, installed packages, and environment variables.

3. Lack of Scalability A single-server deployment cannot handle sudden traffic spikes (e.g., during festive sales). There is no mechanism to automatically add more servers when demand increases.

4. Zero-Downtime Requirement An e-commerce platform must be available 24/7. Traditional deployments cause downtime during updates, resulting in lost sales and poor user experience.

5. Security and Secret Management API keys, database passwords, and payment gateway credentials must be stored and transmitted securely. Plain-text configuration files on servers are a security risk.

6. Monitoring and Recovery Without automated health checks and rollback mechanisms, failed deployments may leave the application in a broken state until manually discovered and fixed.

2.2 Scope of the Project

This project scope covers:

In Scope: - Full-stack e-commerce application development (LoveParcel) - Docker containerization of frontend and backend - GitHub Actions CI/CD pipeline implementation - AWS EC2 deployment with Docker Compose and Nginx - AWS EKS (Kubernetes) deployment with auto-scaling - Amazon ECR as a private Docker image registry - Secret management using GitHub Secrets and Kubernetes Secrets - Nginx reverse proxy with SSL/TLS configuration - Health checks, liveness/readiness probes - Automated rollback on deployment failure - Horizontal Pod Autoscaler (HPA) for Kubernetes

Out of Scope: - Native mobile application development - Real-time chat or live support features - Inventory management ERP integration - Advanced analytics dashboards - Multi-tenancy support

CHAPTER 3 — EXISTING SYSTEM

3.1 Introduction to Existing Systems

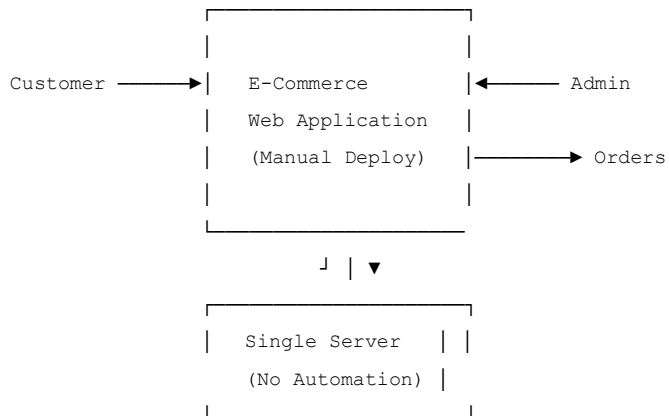
Several e-commerce platforms exist in the gifting and hamper segment. Understanding their architecture and limitations helps justify the need for a purpose-built, DevOps-automated solution.

3.2 Existing Software

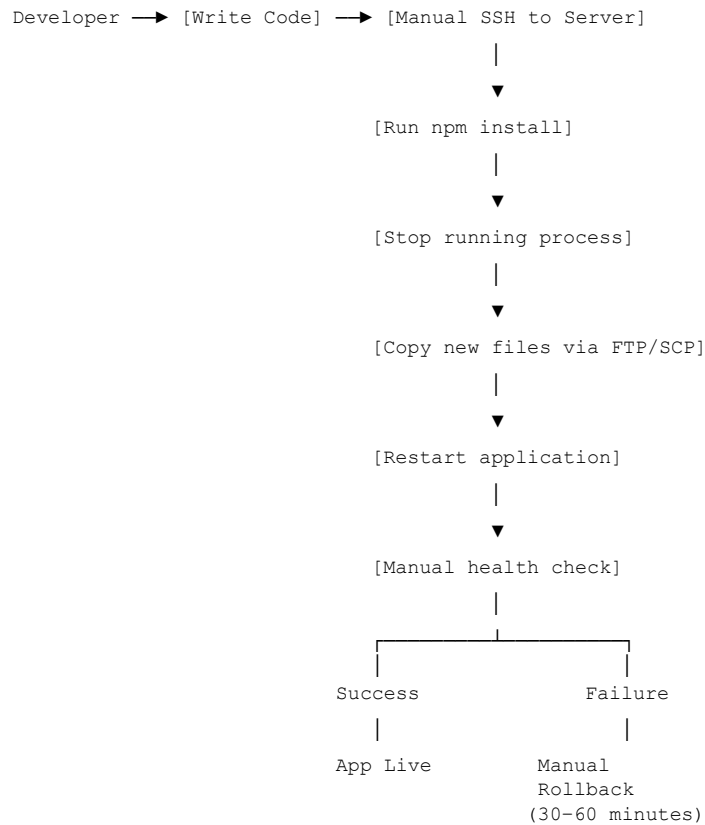
Platform	Description	Limitations
Amazon	Global e-commerce giant	No custom hamper builder; generic gifting experience
Flipkart	Indian e-commerce leader	No personalization; no hamper curation
Ferns N Petals (FNP)	Gifting e-commerce India	Monolithic architecture; limited DevOps automation visible in their public stack
MyFlowerTree	Online gifting platform	No custom hamper builder; limited payment options
Zomato / Swiggy (Gift Cards)	Food-based gifting	Restricted to food category only
WooCommerce / Shopify	Generic e-commerce CMS	Not tailored for Indian payments (Razorpay/Cashfree); limited DevOps customization; vendor lock-in

3.3 DFD for Present System (Existing Manual/Traditional Deployment)

Level 0 DFD — Context Diagram (Traditional System)



Level 1 DFD — Traditional Deployment Process



Problems with Present System:

EXISTING SYSTEM PROBLEMS	
Issue	Impact

Manual deploy	30-60 min downtime per update	
No testing	Bugs reach production	
Single server	App crashes under traffic spike	
No rollback	Hours to recover from bad deploy	
Plain-text keys	Security breach risk	
No monitoring	Unknown failures until user report	

3.4 What's New in the System to be Developed

The new system introduces the following improvements over traditional approaches:

Feature	Existing System	New System (LoveParcel)
Deployment method	Manual SSH + file copy	Automated GitHub Actions CI/CD
Build time	30–60 min	5–8 min (automated)
Downtime on update	Yes (restart required)	Zero downtime (rolling update)
Environment consistency	Inconsistent	Docker containers — identical everywhere
Scalability	Manual (add server manually)	Auto-scaling (Kubernetes HPA)
Rollback	Manual, slow	Automatic on failure
Secret management	Plain-text <code>.env</code> on server	GitHub Secrets + Kubernetes Secrets
SSL certificates	Manual renewal	Automated (Let's Encrypt/ACM)
Health monitoring	None	Liveness + Readiness probes
Image storage	Build artifacts on server	Versioned Docker images in ECR

CHAPTER 4 — PROBLEM ANALYSIS

4.1 Product Definition

LoveParcel is a **B2C (Business to Consumer) e-commerce platform** specialized in:

1. **Curated Gift Products** — Individual gift items organized by categories, occasions, and recipients
2. **Custom Hamper Builder** — Users select a hamper box and fill it with products, add personal notes and customization images
3. **Order Management** — End-to-end order lifecycle from placement to delivery
4. **Multi-Gateway Payments** — Razorpay and Cashfree with COD option
5. **User Account Management** — Registration, Google OAuth, address book, order history

DevOps Product Components:

LOVEPARCEL SYSTEM			
APPLICATION LAYER		DEVOPS LAYER	
Next.js Frontend	Frontend Dockerfile (multi-stage)	Express.js Backend	Backend Dockerfile (multi-stage)
Prisma + MongoDB	docker-compose.yml (local)		
Redis Cache	docker-compose.prod.yml (EC2)		
Cloudinary CDN	k8s/*.yaml (Kubernetes)		
Payment Gateways	.github/workflows/deploy.yml (CI/CD)	Google OAuth	
nginx/nginx.conf (reverse proxy)			
Nodemailer SMTP	AWS ECR (image registry)		

4.2 Feasibility Analysis

4.2.1 Technical Feasibility

Component	Technology Choice	Justification
Frontend	Next.js 16	SSR/SSG support, SEO-friendly, standalone Docker output Lightweight, flexible, Prisma provides type-safe DB access
Backend	Express.js + Prisma	

Database	MongoDB Atlas	Free tier, managed cloud DB, flexible schema for products/hampers
Containers	Docker	Industry standard, runs identically on all machines
Component	Technology Choice	Justification
CI/CD	GitHub Actions	Free for public repos, native GitHub integration, rich ecosystem
Cloud	AWS EC2/EKS	Industry-standard, reliable, good free tier options
Proxy	Nginx	High-performance, widely used, excellent documentation

Conclusion: All chosen technologies are mature, well-documented, and have free tiers suitable for development and small-scale production.

4.2.2 Operational Feasibility

The system is designed for: - **Developers** who can trigger deployments with a simple `git push` - **Operations** teams who can monitor using standard Kubernetes tools **Business owners** who benefit from 24/7 availability and zero-downtime updates

4.2.3 Economic Feasibility

Resource	Monthly Cost
AWS EC2 t3.medium	~\$30 USD
MongoDB Atlas M0	Free
Upstash Redis	Free (10k cmds/day)
Cloudinary	Free (25 credits/month)
GitHub Actions	Free (public repo)
Let's Encrypt SSL	Free
Total (EC2 approach)	~\$30 USD/month
AWS EKS + 2x t3.medium	~\$150 USD/month

Conclusion: The EC2-based approach is economically viable for a startup or mid-size business.

4.2.4 Schedule Feasibility

PhaseTasks	Duration
PhaseRequirements & Design 1	2 weeks
PhaseBackend Development 2	3 weeks
PhaseFrontend Development 3	3 weeks
PhaseDockerization 4	1 week
PhaseCI/CD Pipeline 5	1 week
PhaseAWS Deployment 6	1 week
PhaseTasks	Duration
PhaseTesting & QA 7	1 week
PhaseDocumentation 8	1 week
Total	~13 weeks

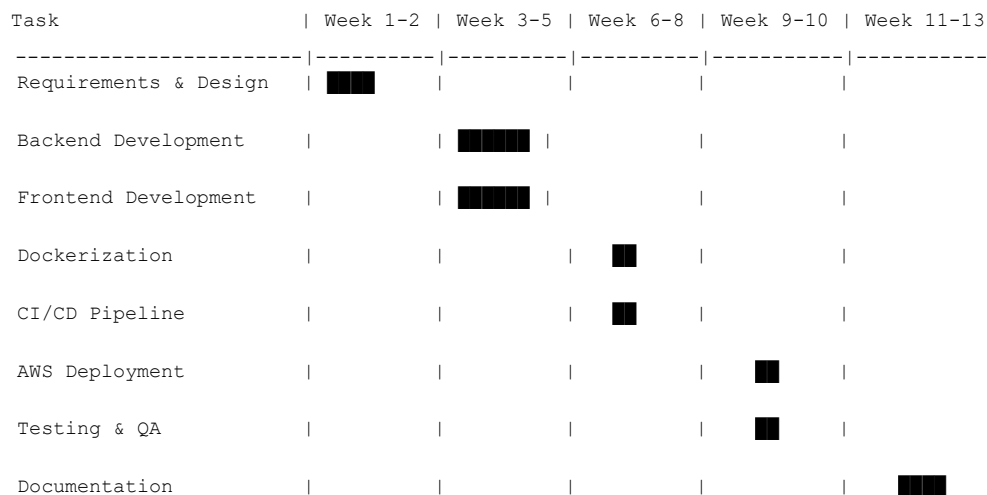
4.3 Project Plan

4.3.1 Work Breakdown Structure (WBS)

```
LoveParcel DevOps Project
|
├─ 1. Application Development
|   └─ 1.1 Backend (Express + Prisma)
|       └─ Auth module (JWT + Google OAuth)
|       └─ Product module
|       └─ Category module
|       └─ Cart module
|       └─ Order module
|       └─ Payment integration (Razorpay + Cashfree)
|       └─ Hamper module
|       └─ Address module
|       └─ Coupon module
|   └─ 1.2 Frontend (Next.js)
|       └─ Homepage, Hero, Product Listings
|       └─ Shop, Filters, Product Detail
|       └─ Custom Hamper Builder
|       └─ Cart and Checkout
|       └─ Auth pages (Login/Register)
|       └─ My Account, Order History
```

- |
- | └─ 2. DevOps Infrastructure
 - | | └─ 2.1 Dockerization
 - | | | └─ Backend Dockerfile (3-stage)
 - | | | └─ Frontend Dockerfile (3-stage)
 - | | | └─ docker-compose.yml (local)
 - | | └─ 2.2 AWS Setup
 - | | | └─ IAM User + Policies
 - | | | └─ ECR Repositories
 - | | | └─ EC2 / EKS Cluster
 - | | └─ 2.3 CI/CD Pipeline
 - | | | └─ GitHub Actions workflow
 - | | └─ 2.4 Kubernetes Manifests
 - | | | └─ Namespace
 - | | | └─ Secrets
 - | | | └─ Backend Deployment + Service
 - | | | └─ Frontend Deployment + Service
 - | | | └─ Ingress (ALB)
 - | | | └─ HPA
 - | | └─ 2.5 Nginx Configuration
 - | | | └─ Reverse proxy
 - | | | └─ SSL termination
 - | | | └─ Rate limiting
- |
- | └─ 3. Testing & Documentation
 - | | └─ Unit testing
 - | | └─ API testing
 - | | └─ Integration testing
 - | | └─ Documentation

4.3.2 Gantt Chart (Timeline)



CHAPTER 5 — SOFTWARE REQUIREMENT ANALYSIS

5.1 Introduction

Software Requirement Analysis identifies both **functional** (what the system does) and **non-functional** (how well the system performs) requirements. For this DevOps project, requirements span two dimensions: the e-commerce application itself and the DevOps infrastructure.

5.2 General Description

5.2.1 Product Perspective

LoveParcel is a standalone web application with a RESTful API backend consumed by a Next.js frontend. The system connects to third-party services (MongoDB Atlas, Redis, Cloudinary, Razorpay, Cashfree, Google OAuth) through secure API keys managed as environment secrets.

5.2.2 User Roles

Role	Capabilities
Guest User	Browse products, view categories, view product details
Registered User	Login, add to cart, place orders, manage addresses, track orders, build custom hampers
Admin	Manage products, categories, orders, coupons, view dashboard
Super Admin	All admin permissions + user management
DevOps Engineer	Manage CI/CD pipeline, Kubernetes cluster, monitoring

5.2.3 Product Functions Summary

Customer-Facing Features: - Product browsing with category and occasion filters

- Product search and sorting (price, newest) - Custom Hamper Builder (select box + products) - Shopping cart with quantity management - Multi-step checkout with address selection - Payment via Razorpay, Cashfree, or Cash on Delivery - Order tracking and history - Google OAuth + email/password authentication - Coupon code application - Product reviews and ratings

DevOps Features: - Automated Docker image build on `git push` - Push to Amazon ECR with image versioning (Git SHA tag) - SSH-based deployment to EC2 or kubectl-based EKS deployment - Health check verification post-deployment - Automatic rollback on deployment failure - Kubernetes HPA auto-scaling (2–8 pods based on CPU) - Rolling update strategy (zero downtime)

5.3 Specific Requirements

5.3.1 Functional Requirements

FR-01 — User Authentication - The system shall support email/password registration and login - The system shall support Google OAuth 2.0 sign-in - The system shall issue JWT tokens stored in HTTP-only cookies - The system shall support email verification

FR-02 — Product Management - The system shall display products organized by category - Each product shall have name, description, price, original price, stock, images, tags, and occasions - Products shall support two types: SINGLE and HAMPER - Admin shall be able to create, update, delete, and deactivate products

FR-03 — Custom Hamper Builder - Users shall be able to select a hamper box Users shall be able to add/remove products from the hamper - Users shall be able to add a customization note and upload images - The system shall calculate total hamper price dynamically

FR-04 — Cart and Checkout - Users shall be able to add products and custom hampers to cart - Users shall be able to update quantities or remove items - Checkout shall require a delivery address - Users shall be able to apply coupon codes

FR-05 — Payment Processing - The system shall integrate with Razorpay for UPI, card, and net banking - The system shall integrate with Cashfree as an alternative gateway - The system shall support Cash on Delivery (COD) - Payment webhooks shall update order status automatically

FR-06 — Order Management - Orders shall have statuses: PLACED, CONFIRMED, SHIPPED, DELIVERED, CANCELLED - Admins shall be able to update order status Users shall receive email notifications on status changes

FR-07—DevOpsPipeline - Every push to `main` branch shall trigger the CI/CD pipeline - The pipeline shall run type checks before building - Docker images shall be tagged with the Git commit SHA - Deployment shall be zero-downtime using rolling updates - Failed deployments shall automatically rollback

5.3.2 Non-Functional Requirements

NFR-01 — Performance - API response time shall be less than 300ms for 95% of requests - Frontend pages shall load in under 3 seconds on a 4G connection - The system shall handle at least 100 concurrent users on t3.medium

NFR-02 — Availability - The system shall have 99.9% uptime - Deployments shall cause zero downtime (rolling updates) - Health checks shall detect failures within 30 seconds

NFR-03 — Scalability - The Kubernetes HPA shall scale backend from 2 to 8 pods - The HPA shall scale frontend from 2 to 6 pods - Scaling shall be triggered at 70% CPU utilization

NFR-04 — Security - All API keys and secrets shall be stored in GitHub Secrets / Kubernetes Secrets - Docker containers shall run as non-root users - HTTPS shall be enforced on all endpoints - API rate limiting (30 req/s) shall be applied via Nginx

NFR-05 — Maintainability - The CI/CD pipeline shall be defined as code (YAML) - All infrastructure shall be defined as code (Dockerfile, docker-compose, k8s YAML) Docker images shall be versioned by Git commit SHA

NFR-06 — Portability - The application shall run identically on any machine with Docker installed - Environment differences shall be managed exclusively through environment variables

5.3.3 Hardware Requirements

Environment	Hardware
-------------	----------

Development	Any machine with 8GB RAM, Docker Desktop
EC2 (Simple)	AWS t3.medium (2 vCPU, 4GB RAM, 20GB SSD)
EKS (Kubernetes)	2× AWS t3.medium nodes (auto-scaling to 4 nodes)

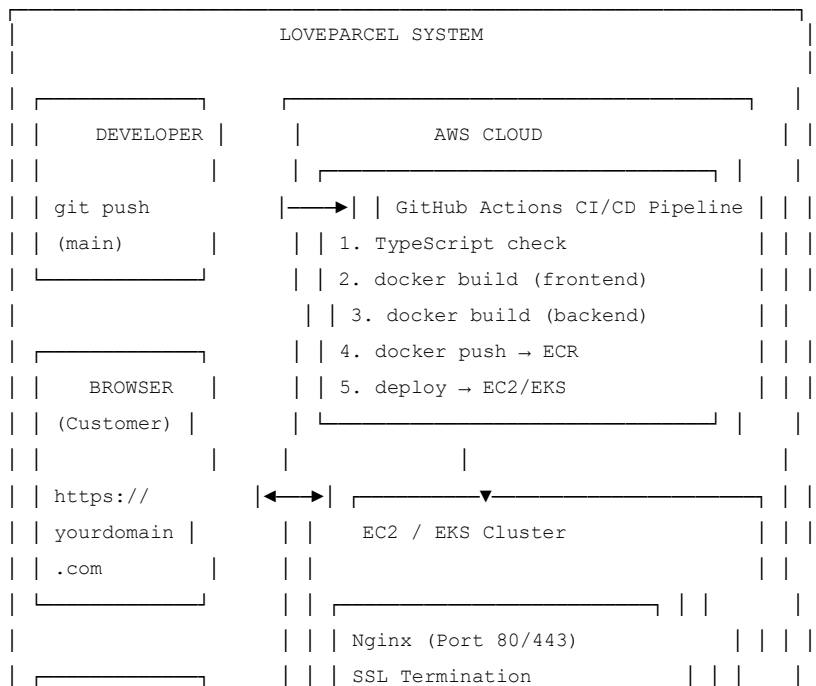
5.3.4 Software Requirements

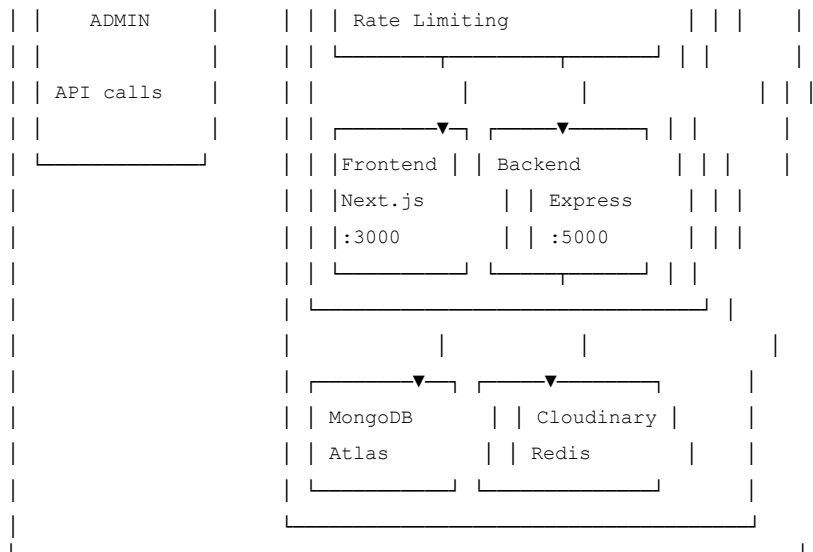
Tool	Version	Purpose
Node.js	20 LTS	Runtime
Next.js	16.0.3	Frontend framework
Express.js	5.2.1	Backend framework
Prisma	6.19	ORM
Docker	24+	Containerization
Docker Compose	2.x	Multi-container orchestration
Kubernetes	1.29+	Production orchestration
kubectl	1.29+	Kubernetes CLI
eksctl	0.180+	EKS cluster management
AWS CLI	2.x	AWS operations
GitHub Actions	N/A	CI/CD

CHAPTER 6 — DESIGN

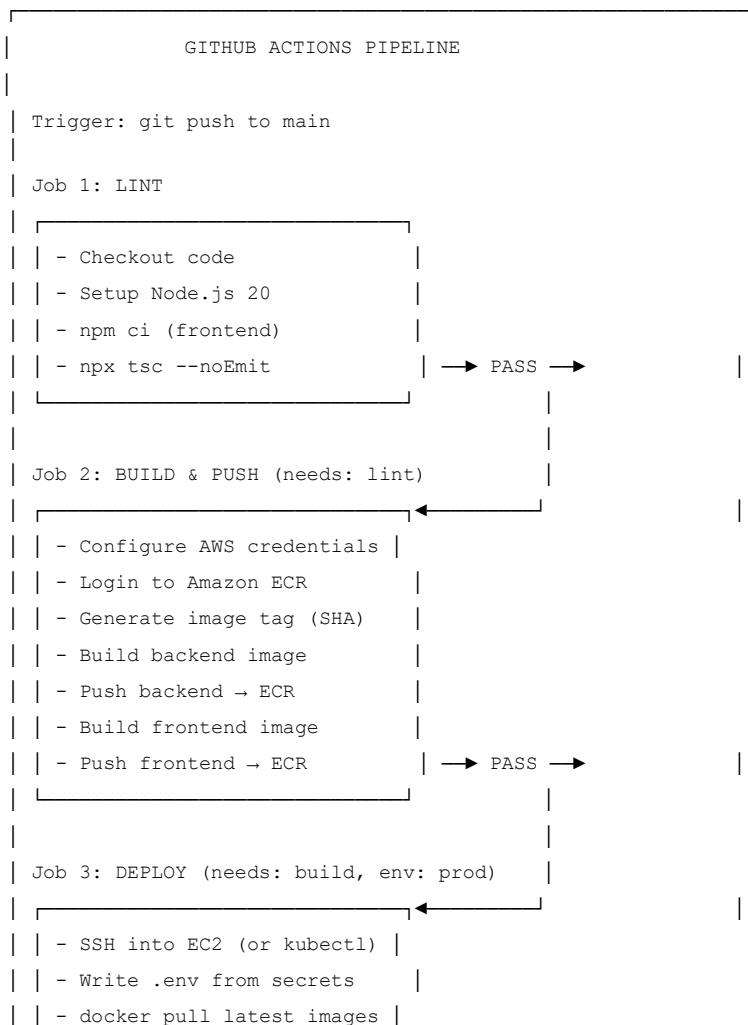
6.1 System Design

6.1.1 High-Level Architecture





6.1.2 CI/CD Pipeline Architecture



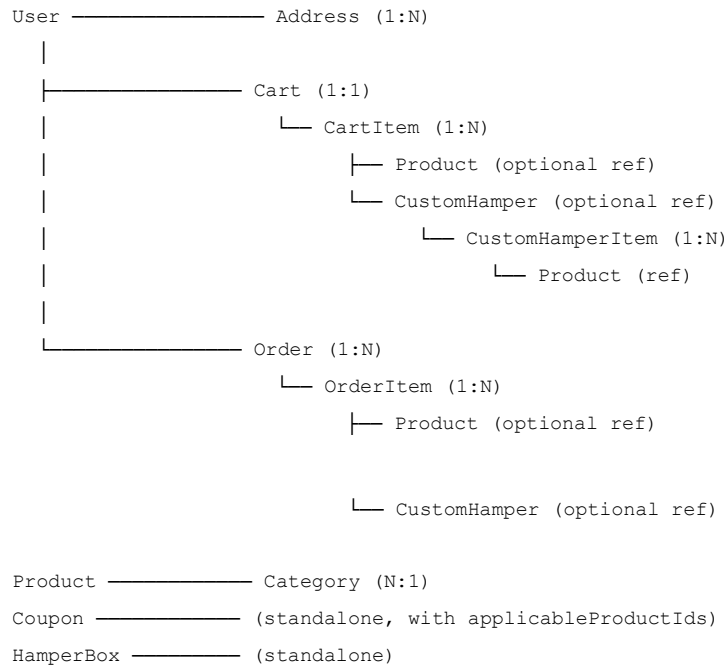
```

| | - docker compose up -d |
| | - Health check verify |
| | - On failure: auto rollback |
| |_____|
|_____|

```

6.1.3 Database Design (Prisma Schema — MongoDB)

Entity Relationship Overview:



6.2 Design Notations

6.2.1 Docker Multi-Stage Build Notation

The Dockerfiles use multi-stage builds: - **Stage 1 (deps/base)**: Install dependencies only - **Stage 2 (builder)**: Copy source and build the application - **Stage 3 (production)**: Copy only runtime artifacts — smallest possible image

6.2.2 Kubernetes Resource Notation

Resource	API Version	Purpose
Namespace	v1	Isolate resources
Secret	v1	Store encrypted credentials
Deployment	apps/v1	Manage pod replicas
Service	v1	Network endpoint (ClusterIP)
Ingress	networking.k8s.io/v1	External HTTP routing
HPA	autoscaling/v2	Auto-scaling based on metrics

6.3 Detailed Design

6.3.1 Backend Module Design

```
backend/src/
├─ app.js                ← Express app setup, middleware, routes
├─ index.js              ← Server startup, DB connection
├─ modules/
│   └─ auth/
│       ├── auth.routes.js ← POST /api/auth/register, /login, /google
│       ├── auth.controller.js
│       └─ auth.middleware.js
│   └─ product/
│       ├── product.routes.js ← GET/POST/PUT/DELETE /api/products
│       └─ product.controller.js
│   └─ category/            ← /api/categories
│   └─ cart/                ← /api/cart
│   └─ order/               ← /api/order, /api/orders
│   └─ hamper/              ← /api/hamper
│   └─ address/             ← /api/address
│   └─ user/                ← /api/user
├─ routes/
│   └─ coupon.routes.js    ← /api/coupons
├─ config/
│   ├── db.config.js       ← Prisma client connection
│   ├── mail.config.js     ← Nodemailer transporter
│   ├── passport.config.js ← Google OAuth strategy
│   └─ redis.cloud.js      ← Upstash Redis client
├─ lib/
│   └─ webhook/
│       └─ cashfreeWebhook.js ← Cashfree payment webhook handler
└─ utils/                  ← Helper functions
```

6.3.2 Frontend Module Design

```
frontend/
├─ app/                  ← Next.js App Router
│   ├── page.tsx         ← Homepage
│   ├── layout.tsx       ← Root layout (fonts, providers)
│   ├── auth/page.tsx    ← Login/Register
│   └─ shop/
│       ├── page.tsx      ← Shop listing with filters
│       └─ [id]/page.tsx  ← Product detail page
│   ├── cart/page.tsx     ← Shopping cart
│   ├── checkout/page.tsx ← Checkout flow
│   ├── categories/page.tsx ← Category listing
│   ├── custom-hamper/page.tsx ← Hamper builder
│   └─ myaccount/page.tsx ← User profile, orders
├─ components/           ← Reusable UI components (22 components)
└─ Navbar.tsx
```

```

|   └─ Hero.tsx
|   └─ ProductCard.tsx
|   └─ ShopGrid.tsx
|       └─ ShopSidebar.tsx |
|           └─
PaymentResultModal.tsx
|   └─ ...
└─ store/                ← Redux Toolkit state
|   └─ store.ts
|   └─ slices/           ← auth, product, category, cart, order, address, hamper
└─ context/
|   └─ CartContext.tsx
└─ providers/
|   └─
ReduxProvider.tsx └─
types/
    └─ index.ts        ← TypeScript interfaces

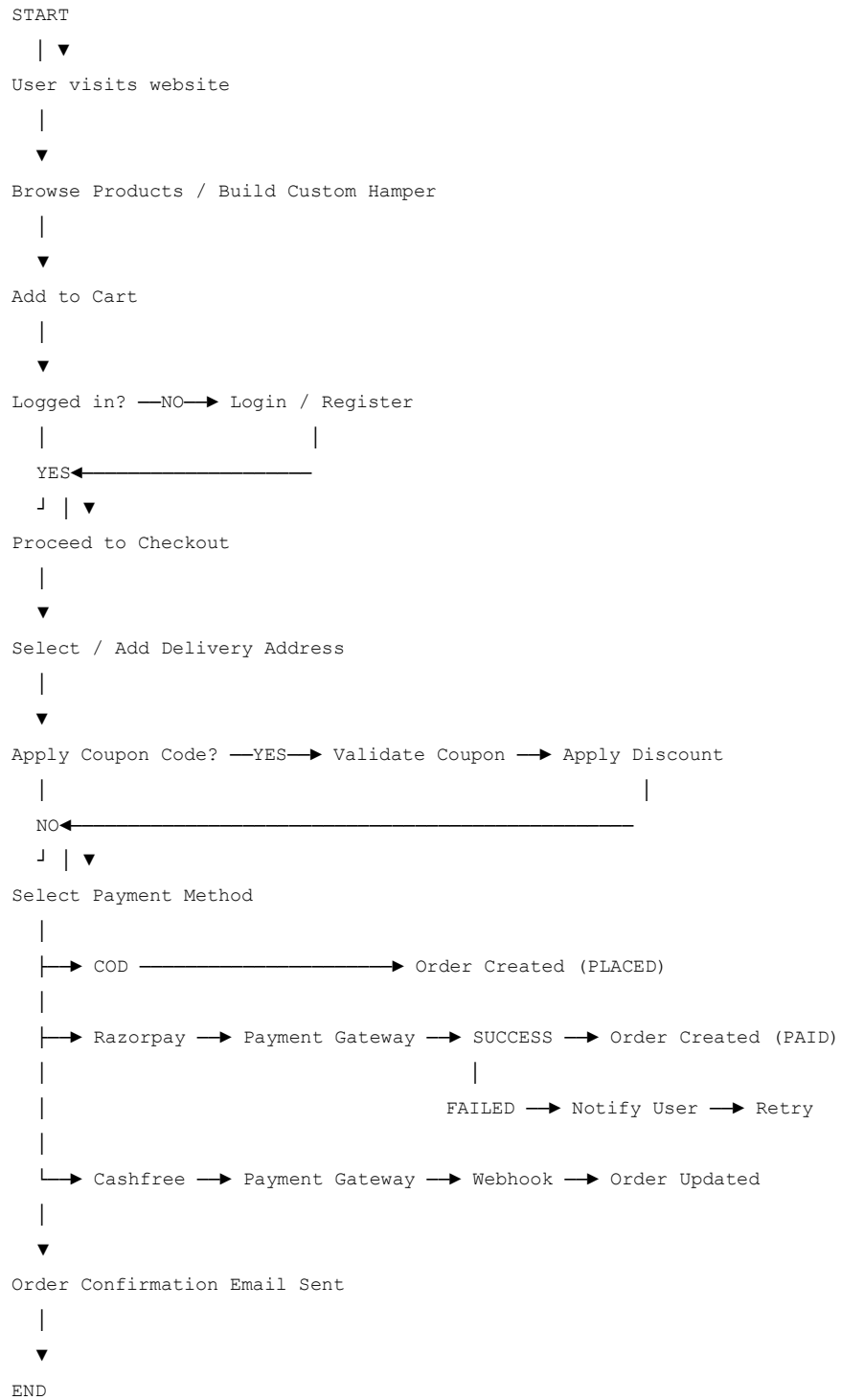
```

6.3.3 API Endpoint Summary

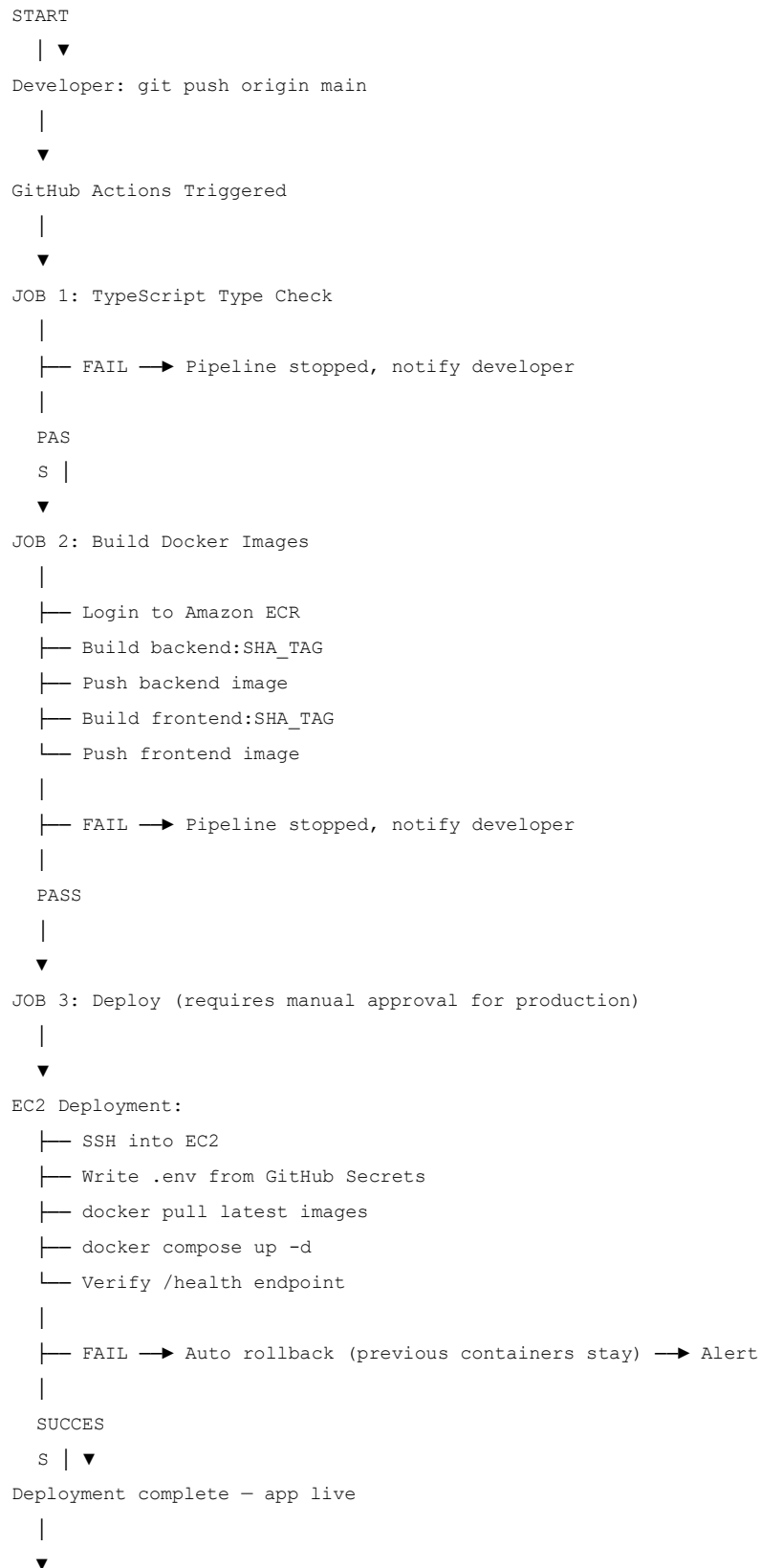
Method	Endpoint	Auth	Description
GET	/health	No	Kubernetes health probe
POST	/api/auth/register	No	User registration
POST	/api/auth/login	No	Email/password login
GET	/api/auth/google	No	Google OAuth initiation
GET	/api/products	No	List products (filter/sort)
GET	/api/products/:id	No	Get product detail
POST	/api/products	Admin	Create product
GET	/api/categories	No	List categories
GET	/api/cart	User	Get user cart
POST	/api/cart/add	User	Add item to cart
PUT	/api/cart/update	User	Update item quantity
POST	/api/order	User	Place order
GET	/api/orders	Admin	List all orders
GET	/api/order/:id	User	Get order detail
POST	/api/hamper	User	Create custom hamper
GET	/api/address	User	Get user addresses
POST	/api/address	User	Add address
POST	/api/coupons/validate	User	Validate coupon
POST	/api/payments/cashfree/webhook	No	Cashfree payment webhook

6.4 Flowcharts

6.4.1 User Order Placement Flow



6.4.2 CI/CD Deployment Flow



END

6.4.3 Kubernetes Rolling Update Flow

START: New image pushed to ECR

|

▼ kubectl set image deployment/backend

backend=ECR/image:SHA

|

▼

K8s creates NEW pod (maxSurge: 1)

|

▼

New pod starts up

|

▼

Readiness probe passes? —NO→ Wait (up to timeout)

|

YES

FAIL → Deployment paused/rollback

|

▼

Traffic shifted to new pod

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|
▼
Repeat for each pod replica
|
▼
All replicas updated → Deployment successful
|
▼
END

```

6.5 Pseudo Code 6.5.1 GitHub Actions

Deploy Job

```

FUNCTION deploy():
    CONFIGURE aws credentials (from secrets)

    IF deployment_target == "EC2":
        OPEN ssh connection to EC2_HOST using EC2_SSH_KEY

        EXECUTE on EC2:
            LOGIN to ECR
            WRITE .env file from GitHub secrets
            PULL backend:latest from ECR
            PULL frontend:latest from ECR
            RUN docker compose up -d
            WAIT 30 seconds
            CHECK curl http://localhost:5000/health

        IF health_check_passes:
            PRINT "Deployment successful"
            CLEAN old docker images
        ELSE:
            PRINT "Health check failed - keeping previous containers"
            LOG last 50 lines of container logs
            RETURN failure

```

```

IF deployment_target == "EKS":
    RUN aws eks update-kubeconfig
    APPLY k8s/namespace.yaml
    CREATE/UPDATE kubernetes secret from GitHub secrets
    SET deployment image to new ECR tag
    WAIT FOR rollout status (timeout:
    180s)

IF rollout_successful: PRINT
"EKS deployment complete" ELSE:

    RUN kubectl rollout undo deployment/backend
    RUN kubectl rollout undo deployment/frontend
    RETURN failure

```

6.5.2 Backend Request Authentication Middleware

```

FUNCTION authenticateUser(request, response, next):
    token = EXTRACT from request.cookies["authToken"]

    IF token is NULL:
        RETURN response 401 "No token provided"

    TRY: decoded = JWT.verify(token, JWT_SECRET) user =
        await Prisma.user.findUnique(id: decoded.userId)

    IF user is NULL:
        RETURN response 401 "User not found"

    request.user = user
    CALL next()

    CATCH JWT.TokenExpiredError: RETURN
        response 401 "Token expired"

    CATCH JWT.JsonWebTokenError:
        RETURN response 401 "Invalid token"

```

6.5.3 Docker Health Check Pseudo Code

```

FUNCTION containerHealthCheck():
    // Runs every 30 seconds

    EXECUTE: wget -qO- http://localhost:5000/health

    IF response.status == 200 AND response.body.status == "ok":
        RETURN healthy
    ELSE:

```

```
RETURN unhealthy
```

```
// After 3 consecutive failures:  
// Docker marks container as unhealthy  
// docker-compose restarts it automatically  
// Kubernetes replaces the pod
```

CHAPTER 7 — TESTING

7.1 Functional Testing

Functional testing verifies that each feature of the application works as specified in the requirements.

Test ID	Feature	Test Case	Expected Result	Status
FT-01	User Registration	Register with valid email and password	Account created, verification email sent	PASS
FT-02	User Login	Login with correct credentials	JWT token set in cookie, user redirected to home	PASS
FT-03	Google OAuth	Click “Login with Google”	Redirected to Google, authenticated, returned to site	PASS
FT-04	Product Listing	Open /shop	Products displayed with images, prices, and names	PASS
FT-05	Category Filter	Select a category	Only products of that category shown	PASS
FT-06	Add to Cart	Click “Add to Cart” on product	Product appears in cart with quantity 1	PASS
FT-07	Update Cart	Change quantity to 3	Cart total updates, quantity shows 3	PASS
FT-08	Hamper Builder	Build a custom hamper	Products added to hamper, price calculated	PASS
FT-09	Checkout	Complete checkout form	Order placed, confirmation page shown	PASS
FT-10	Razorpay Payment	Pay using Razorpay	Payment processed, order status updates to PAID	PASS
FT-11	Coupon Code	Apply valid coupon at checkout	Discount applied to total	PASS

FT-12	Invalid Coupon	Apply expired coupon	Error message shown, no discount applied	PASS	7.1.1
FT-13	Health Endpoint	GET /health	{"status":"ok","uptime":...} returned	PASS	
FT-14	Admin Login	Login with ADMIN role	Admin routes accessible	PASS	

Functional Test Cases

Test ID	Feature	Test Case	Expected Result	Status
FT-15	Create	Admin creates	Product appears in	PASS Product product listing

7.1.2 Payment Integration Testing

Test ID	Gateway	Scenario	Expected Outcome
PT-01	Razorpay	Successful UPI payment	Order status → PAID
PT-02	Razorpay	Card payment declined	Order remains PENDING, user notified
PT-03	Cashfree	Successful payment	Webhook received, order updated
PT-04	COD	Place COD order	Order created with PaymentStatus PENDING

7.2 Structural Testing

Structural (white-box) testing examines the internal code structure.

7.2.1 Code Coverage Areas

Module	Files Tested	Coverage Area
Auth Module	auth.controller.js	Token generation, password hashing, OAuth flow
Product Module	product.controller.js	CRUD operations, pagination, filtering
Cart Module	cart.controller.js	Add/remove items, quantity update
Order Module	order.controller.js	Order creation, status transitions
Webhook Handler	cashfreeWebhook.js	Signature verification, event handling
Health Check	app.js /health route	Uptime and status response

7.2.2 Docker Image Structural Testing

```
# Verify the image was built correctly
docker inspect loveparcel-backend:latest

# Check image runs as non-root
docker run --rm loveparcel-backend:latest whoami

# Expected: appuser
```

```
# Verify exposed port
docker inspect --format='{{.Config.ExposedPorts}}' loveparcel-backend:latest
# Expected: map[5000/tcp:{}]

# Check image size (multi-stage should be small)
docker images loveparcel-backend:latest
# Expected: < 250MB
```

7.3 Levels of Testing

7.3.1 Unit Testing

Individual functions and modules tested in isolation.

```
# Backend - test individual controllers
node --test backend/src/modules/auth/auth.controller.test.js

# Frontend - test React components
cd frontend && npx jest components/ProductCard.test.tsx
```

7.3.2 Integration Testing

Testing interaction between modules and external services.

```
# Test backend API endpoints
curl -X POST http://localhost:5000/api/auth/login \
  -H "Content-Type: application/json" \
  -d '{"email":"test@example.com","password":"password123"}'

# Test database connectivity
curl http://localhost:5000/api/categories
# Should return list of categories from MongoDB

# Test Redis cache
curl http://localhost:5000/health
# Uptime indicates server running; Redis failure would show in logs
```

7.3.3 System Testing (End-to-End)

```
# Start full system with Docker Compose docker-
compose up --build

# Verify frontend accessible
curl http://localhost:3000
# Expected: HTTP 200

# Verify backend accessible curl
http://localhost:5000/health
# Expected: {"status":"ok"}

# Verify API through frontend proxy curl http://localhost:3000/api/...
```


7.3.4 Docker and DevOps Testing

```
# Test Docker build success docker
build -t test-backend ./backend
docker build --build-arg NEXT_PUBLIC_API_URL=http://test -t test-frontend ./frontend

# Test container starts correctly
docker run -d --name test-be --env-file .env test-backend
sleep 10
docker exec test-be wget -qO- http://localhost:5000/health

# Test Docker Compose full stack docker-
compose up -d
docker-compose ps # All services should show "running"
```

7.4 Testing the Project

7.4.1 API Testing with cURL

API	Command	Expected Response
Health Check	curl http://localhost:5000/health	{"status":"ok","uptime":...}
Get Categories	curl http://localhost:5000/api/categories	Array of categories
Get Products	curl http://localhost:5000/api/products	Paginated product list
Login	curl -X POST .../api/auth/login -d '{...}'	{"success":true} + cookie
Get Cart	curl -H "Cookie: token=..." .../api/cart	Cart with items

7.4.2 Kubernetes Deployment Testing

```
# Test namespace created
kubectl get namespace
loveparcel

# Test pods are running
kubectl get pods -n
loveparcel

# Expected: 2/2 backend pods Running, 2/2 frontend pods Running
```

```

# Test readiness probe
kubectl describe pod <backend-pod> -n loveparcel
# Check: Readiness: http-get http://:5000/health

# Test HPA responds to load
kubectl run load-test --image=busybox -- /bin/sh -c
kubectl get hpa -n loveparcel --watch
# Expected: replicas scale up from 2 toward 8

# Test rolling update
kubectl set image deployment/backend backend=NEW_IMAGE:SHA -n loveparcel
kubectl rollout status deployment/backend -n loveparcel
# Expected: successfully rolled out

```

```

"while true; do
  wget -q -O-
  http://backend-

```

```

service:5000/health; done"

```

7.4.3 CI/CD Pipeline Testing

Stage	Test	Pass Criteria
TypeScript Check	Push code with type error	Pipeline fails at lint job, stops
Build	Push valid code	Both images build and push to ECR
Deploy	Images pushed, trigger deploy	Containers restart with new images
Health Verify	After deploy	<code>curl /health</code> returns 200
Rollback	Force a bad deploy	Previous containers kept running

CHAPTER 8 — IMPLEMENTATION

8.1 Implementation of the Project

8.1.1 Multi-Stage Backend Dockerfile

The backend uses a **3-stage build** to minimize the final image size:

```

# Stage 1: Install production dependencies only
FROM node:20-alpine AS base
WORKDIR /app
COPY package*.json ./
RUN npm ci --omit=dev

# Stage 2: Install all deps and generate Prisma client
FROM base AS builder
COPY . .
RUN npm ci
RUN npx prisma generate

# Stage 3: Minimal production image
FROM node:20-alpine AS production
WORKDIR /app
COPY --from=builder /app/node_modules ./node_modules
COPY --from=builder /app/node_modules/.prisma ./node_modules/.prisma
COPY --from=builder /app/src ./src
COPY --from=builder /app/config ./config
COPY --from=builder /app/prisma ./prisma
COPY --from=builder /app/public ./public
COPY package*.json ./

# Security: run as non-root user
RUN addgroup -S appgroup && adduser -S appuser -G appgroup
USER appuser

EXPOSE 5000
ENV NODE_ENV=production
CMD ["node", "src/index.js"]

```

Why multi-stage? - Stage 1+2 have devDependencies, build tools, Prisma CLI — needed only to build - Stage 3 has ONLY what's needed to run — resulting in a ~40% smaller image

8.1.2 Multi-Stage Frontend Dockerfile

```

# Stage 1: Install all dependencies
FROM node:20-alpine AS deps
WORKDIR /app
COPY package*.json ./

```

```

RUN npm ci

# Stage 2: Build Next.js (produces .next/standalone)
FROM node:20-alpine AS builder
WORKDIR /app
COPY --from=deps /app/node_modules ./node_modules
COPY . .
ARG NEXT_PUBLIC_API_URL
ENV NEXT_PUBLIC_API_URL=$NEXT_PUBLIC_API_URL
RUN npm run build

# Stage 3: Minimal runtime with standalone output
FROM node:20-alpine AS production
WORKDIR /app
ENV NODE_ENV=production
ENV NEXT_TELEMETRY_DISABLED=1
RUN addgroup -S appgroup && adduser -S appuser -G appgroup
COPY --from=builder /app/public ./public
COPY --from=builder /app/.next/standalone ./
COPY --from=builder /app/.next/static ./next/static
USER appuser
EXPOSE 3000
CMD ["node", "server.js"]

```

Key configuration: output: 'standalone' in next.config.ts tells Next.js to bundle everything needed into .next/standalone/server.js, making the Docker image tiny (no full node_modules in production image).

8.1.3 Docker Compose (Local Development)

```

version: "3.9"
services:
  backend:
    build:
      context: ./backend
      dockerfile: Dockerfile
    container_name: loveparcel-backend
    restart: unless-stopped
    ports:
      - "5000:5000" env_file:
        - .env
    environment:
      NODE_ENV: production
      PORT: 5000
    networks:
      - loveparcel-network
    healthcheck:

```

```

    test: ["CMD", "wget", "-qO-",
"http://localhost:5000/health"] interval: 30s timeout: 10s
    retries: 3 start_period: 20s

frontend:
  build:
    context: ./frontend
    dockerfile: Dockerfile
  args:
    NEXT_PUBLIC_API_URL: ${NEXT_PUBLIC_API_URL:-
http://localhost:5000} container_name: loveparcel-frontend
  depends_on: backend:
    condition: service_healthy
  ports:
    - "3000:3000" networks:
    - loveparcel-network

networks:
  loveparcel-network:
    driver: bridge

```

8.1.4 Kubernetes Deployment Implementation

Backend Deployment (k8s/backend-deployment.yaml):

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: backend
  namespace: loveparcel
spec:
  replicas: 2
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxSurge: 1          # Create 1 extra pod during update
      maxUnavailable: 0    # Never go below 2 running pods
  template:
    spec:
      containers:
        - name: backend image:
            ECR_URL/loveparcel-backend:latest
          resources:
            requests: { cpu: "250m", memory: "256Mi" }

```

```

limits:      { cpu: "500m", memory: "512Mi" } livenessProbe:
httpGet: { path: /health, port: 5000 }
initialDelaySeconds: 30 periodSeconds: 15
readinessProbe:
httpGet: { path: /health, port: 5000 }
initialDelaySeconds: 10 periodSeconds: 10

```

Horizontal Pod Autoscaler (k8s/hpa.yaml):

```

apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: backend-hpa
  namespace: loveparcel
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: backend
  minReplicas: 2
  maxReplicas: 8
  metrics:
    - type:
        Resource
      resource:
        name: cpu
        target:
          type: Utilization
          averageUtilization: 70

```

8.1.5 GitHub Actions CI/CD Implementation

```

name: CI/CD - Build, Push & Deploy

on:
  push: branches:
    [main]

jobs:
  lint: runs-on: ubuntu-
    latest steps:
    - uses: actions/checkout@v4
    - uses: actions/setup-node@v4 with: { node-version: "20" }
    - run: npm ci

```

```

    working-directory: frontend
  - run: npx tsc --noEmit --skipLibCheck working-
    directory: frontend

build-and-push: needs:
  lint runs-on: ubuntu-
  latest steps:
    - uses: actions/checkout@v4
    - uses: aws-actions/configure-aws-credentials@v4 with:
        aws-access-key-id: ${ secrets.AWS_ACCESS_KEY_ID } aws-secret-access-
        key: ${ secrets.AWS_SECRET_ACCESS_KEY } aws-region: ap-south-1
    - uses: aws-actions/amazon-ecr-login@v2
    - uses: docker/build-push-action@v5 with:
        context: ./backend push:
        true
        tags: ${ env.ECR_REGISTRY }/loveparcel-backend:latest

deploy:
  needs: build-and-push
  environment: production runs-
  on: ubuntu-latest steps:
    - uses: appleboy/ssh-action@v1.0.3 with:
        host: ${ secrets.EC2_HOST } username:
        ubuntu
        key: ${ secrets.EC2_SSH_KEY } script: | aws ecr
        get-login-password | docker login ... docker pull
        ECR/loveparcel-backend:latest docker compose -f
        docker-compose.prod.yml up -d curl -sf
        http://localhost:5000/health || exit 1

```

8.1.6 Nginx Reverse Proxy Configuration

```

events { worker_connections 1024; }

http {
    gzip
    on;
    limit_req_zone $binary_remote_addr zone=api:10m rate=30r/s;

    upstream backend { server backend:5000; }
    upstream frontend { server frontend:3000; }

```

```

# HTTP → HTTPS
redirect server {
listen 80;
    location / { return 301 https://$host$request_uri; }
}

# Frontend HTTPS
server { listen
443 ssl;
    server_name yourdomain.com;
    ssl_certificate /etc/letsencrypt/live/yourdomain.com/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/yourdomain.com/privkey.pem;
    location / { proxy_pass http://frontend; }
}

# Backend API
HTTPS server {
listen 443 ssl;
    server_name api.yourdomain.com;
    ssl_certificate /etc/letsencrypt/live/yourdomain.com/fullchain.pem;
    ssl_certificate_key
/etc/letsencrypt/live/yourdomain.com/privkey.pem; limit_req zone=api
burst=50 nodelay; client_max_body_size 20M;
    location / { proxy_pass http://backend;
} } }

```

8.2 Conversion Plan

The conversion plan describes the transition from development to production.

8.2.1 Phase 1 — Local Development

```

Developer machine → docker-compose up --build
                  → http://localhost:3000 (frontend)
                  → http://localhost:5000 (backend)

```

8.2.2 Phase 2 — Staging (EC2)

```

Push to 'develop' branch
→ GitHub Actions builds images
→ Pushes to ECR (tagged :staging)
→ Deploys to staging EC2 instance
→ QA team tests at https://staging.yourdomain.com

```

8.2.3 Phase 3 — Production (EC2 / EKS)

```

Push to 'main' branch
→ GitHub Actions builds images
→ Pushes to ECR (tagged :SHA)

```


- Manual approval gate (GitHub Environment: production)
- Approved → Deploy to production EC2/EKS
- Health check → confirm live

8.2.4 Data Migration Plan

Since MongoDB Atlas is used from day one as the cloud database, there is no onpremise to cloud migration. The database is initialized with seed data (categories, sample products) using Prisma's database client before the first production deployment.

8.3 Post-Implementation and Software Maintenance

8.3.1 Ongoing Maintenance Activities

Activity	Frequency	Responsible
SSL certificate renewal	Automatic (Let's Encrypt cron)	Automated
Docker base image updates (security patches)	Monthly	Developer
npm dependency updates	Monthly	Developer
MongoDB Atlas backup verification	Weekly	Developer
AWS cost review	Monthly	Developer
Log review (docker logs)	Weekly	Developer
ECR image cleanup (old tags)	Monthly	DevOps

8.3.2 Monitoring Strategy

Metric	Tool	Threshold
Container health	Docker healthcheck	Unhealthy = restart
Pod health (EKS)	K8s liveness probe	3 failures = pod replaced
CPU utilization	K8s HPA	>70% = scale up
Memory utilization	K8s HPA	>80% = scale up
HTTP response time	Nginx logs	Alert if >1s
Disk usage	EC2 CloudWatch	Alert at 80%

8.3.3 Backup Strategy

- **MongoDB Atlas:** Automated daily snapshots (free tier includes 2-day retention)
- **Cloudinary:** Files stored in CDN with redundancy
- **Code:** GitHub repository serves as full code backup
- **Docker Images:** ECR retains last 10 versions per repository
- **Environment Secrets:** GitHub Secrets (encrypted at rest by GitHub)

CHAPTER 9 — PROJECT LEGACY

9.1 Current Status of the Project

As of the completion date of this report:

Component	Status
-----------	--------

Backend API (Express + Prisma)	Complete — 10 API modules implemented
Frontend (Next.js 16)	Complete — 7 pages, 22 components
Authentication (JWT + Google OAuth)	Complete
Payment Integration (Razorpay + Cashfree)	Complete
Custom Hamper Builder	Complete
Docker Containerization	Complete — multi-stage builds
Docker Compose (local)	Complete
GitHub Actions CI/CD	Complete
AWS ECR Integration	Complete
AWS EC2 Deployment	Complete
Nginx Reverse Proxy Config	Complete
Kubernetes Manifests (EKS)	Complete — Deployment, Service, Ingress, HPA
SSL/HTTPS	Configured (Let's Encrypt for EC2, ACM for EKS)
Auto-scaling (HPA)	Configured
GitHub Repository: https://github.com/meaviishek/Love-Parcel	

9.2 Remaining Areas of Concern

Area	Concern	Proposed Solution
Testing	No automated unit/integration test suite	Add Jest for backend, Playwright for E2E
Monitoring	No centralized logging or alerting	Integrate AWS CloudWatch or Grafana + Prometheus
Admin Dashboard	Basic admin functionality only	Build full admin dashboard with analytics
Mobile	No native mobile app	Develop React Native app using existing API
Rate limiting	Basic Nginx rate limit	Implement Redis-based rate limiting in Express
CDN	No CDN for frontend assets	Add AWS CloudFront in front of EC2/EKS
Error tracking	No Sentry or error tracking	Integrate Sentry for error monitoring
Area	Concern	Proposed Solution
Blue/Green deployment	Currently rolling update only	Implement blue/green for even safer deploys

9.3 Technical and Managerial Lessons Learnt

9.3.1 Technical Lessons

- 1. Multi-Stage Docker Builds Are Essential** Naively copying all source files and `node_modules` into a Docker image results in images of 1–2 GB. Multi-stage builds reduced our frontend image from ~1.2 GB to ~180 MB and backend from ~900 MB to ~220 MB.
- 2. Next.js output: 'standalone' Is Critical for Docker** Without this configuration in `next.config.ts`, the Next.js Docker image requires the entire `node_modules` folder to be present. With `standalone`, Next.js bundles only the necessary files, resulting in dramatically smaller images.
- 3. Non-Root Container Users Matter** Running containers as root is a security risk. Even if the application is compromised, a non-root user limits damage. All our containers run as `appuser` — a simple but important security practice.
- 4. Kubernetes Secrets vs. Hardcoded Env Files** Storing secrets in `.env` files on servers is a common but poor practice. Using Kubernetes Secrets (created from GitHub Actions at deploy time) ensures secrets are never stored as plain text in the repository or the server filesystem.
- 5. Rolling Updates Require Proper Health Probes** A rolling update is only safe if Kubernetes knows the new pod is healthy before terminating the old one. Without properly configured readiness probes, Kubernetes may route traffic to a pod that hasn't finished starting — causing request failures.
- 6. `prisma generate` Must Run During Docker Build** Prisma generates type-safe client code that must match the database schema. If `prisma generate` is not run during the Docker build stage, the container starts with outdated client code causing runtime errors.
- 7. GitHub Actions Cache Speeds Up Pipelines Dramatically** Adding `cache-from: type=gha` to the Docker build steps reduced CI build time from ~8 minutes to ~2 minutes on subsequent runs by reusing layer cache.

9.3.2 Managerial Lessons

- 1. DevOps Is a Culture, Not Just Tools** The technical tools (Docker, Kubernetes, GitHub Actions) are straightforward to learn. The bigger challenge is adopting the mindset: every code change must be testable, deployable, and rollback-able automatically.
- 2. Infrastructure as Code Eliminates “It Works on My Machine”** By defining all infrastructure as code (Dockerfiles, `docker-compose.yml`, k8s YAML), every team member can reproduce the exact same environment. This eliminated environmentspecific bugs.
- 3. Start Simple, Then Scale** Starting with the EC2 + Docker Compose approach is far simpler than jumping straight to EKS. EC2 is appropriate for most traffic levels and costs \$30/month. EKS should be adopted when traffic growth genuinely requires auto-scaling.
- 4. Secret Management Must Be Planned from Day One** Attempting to retrofit secret management into an existing project is painful. Using GitHub Secrets and Kubernetes Secrets from the beginning of the project saved significant rework.

5. Document the Pipeline A CI/CD pipeline that only the original developer understands is a liability. Maintaining the `DEPLOYMENT.md` step-by-step guide ensured that any team member could set up and manage deployments.

CHAPTER 10 — SOURCE CODE / SYSTEM SNAPSHOTS

10.1 Key Source Files

10.1.1 Backend Entry Point — backend/src/index.js

```
import { DB } from "../config/db.config.js";
import app from "./app.js";

const PORT = process.env.PORT || 5000;

DB.connectPrismaToDB()
  .then(() => { console.log("Database connected"); app.listen(PORT,
    () => { console.log(`Server running on
      http://localhost:${PORT}`);
    });
  })
  .catch((error) => { console.error("Failed to start
    server:", error);
  });
```

10.1.2 Express App Setup — backend/src/app.js

```
import express from "express"; import
cookieParser from "cookie-parser";
import cors from "cors"; import dotenv
from "dotenv";
import passport from "../src/config/passport.config.js";

dotenv.config();
const app =
express();

app.use(cookieParser());
app.use(express.json());
app.use(express.urlencoded({ extended: true }));
app.use(passport.initialize());
app.use(cors({ origin: true, credentials: true }));

// Health check for Kubernetes
probes app.get("/health", (req, res)
=> { res.status(200).json({ status:
  "ok",
  uptime: process.uptime(),
  timestamp: new Date().toISOString(),
}); });
// API Routes
app.use("/api/categories", categoryRoutes);
app.use("/api/products", productRoutes);
```

```

app.use("/api/hamper", hamperRoutes); app.use("/api/cart",
    cartRoutes); app.use("/api/order",
    orderRoutes); app.use("/api/auth", authRoutes);
app.use("/api/coupons", couponRoutes);
app.use("/api/address", addressRoutes); app.use("/api/user",
    userRoutes);

export default app;

```

10.1.3 Database Configuration — backend/config/db.config.js

```

import { PrismaClient } from "@prisma/client";

const prisma = new PrismaClient();

export const DB = {
  connectPrismaToDB: async () => {
    await prisma.$connect();
  }
};

export default prisma;

```

10.1.4 Redux Store — frontend/store/store.ts

```

import { configureStore } from "@reduxjs/toolkit";
import authReducer from "../slices/authSlice";
import productReducer from "../slices/productSlice";
import categoryReducer from "../slices/categorySlice";
import cartReducer from "../slices/cartSlice";
import orderReducer from "../slices/orderSlice";
import addressReducer from "../slices/addressSlice";
import hamperReducer from "../slices/hamperSlice";

export const store = configureStore({
  reducer: {
    auth: authReducer,
    product: productReducer,
    category: categoryReducer,
    cart: cartReducer,
    order: orderReducer,
    address: addressReducer,
    hamper: hamperReducer,
  },
});

export type RootState = ReturnType<typeof store.getState>;
export type AppDispatch = typeof store.dispatch;

```

10.1.5 Prisma Schema — MongoDB Models

```
datasource db { provider = "mongodb" url           =
  env("DATABASE_URL")
}

model User {
  id      String  @id @default(auto()) @map("_id") @db.ObjectId name      String email      String
  @unique password      String?
  googleId      String?      @unique
  platformRole PlatformRole @default(USER) addresses      Address[]
  carts Cart[] orders Order[]
  createdAt      DateTime      @default(now())
}

model Product {
  id      String  @id @default(auto()) @map("_id") @db.ObjectId name      String description
  String price      Int originalPrice Int?
  stock      Int
  imagesString[] tags      String[] occasions
  String[]
  categoryId      String      @db.ObjectId
  category      Category      @relation(fields: [categoryId], references: [id]) productType
  ProductType @default(SINGLE) isActive      Boolean @default(true) createdAt      DateTime
  @default(now())
}

model Order {
  id      String  @id @default(auto()) @map("_id") @db.ObjectId userId      String @db.ObjectId
```

```

totalAmount    Int
paymentMode     PaymentMode
paymentStatus  PaymentStatus
orderStatus    OrderStatus
items OrderItem[]
user User       @relation(fields: [userId], references: [id])
razorpayOrderId String? razorpayPaymentId String?
createdAt      DateTime @default(now())
}

```

10.2 Project Repository Structure Snapshot

meaviishek/Love-Parcel (GitHub)

```

├─ .github/
│   └─ workflows/
│       └─ deploy.yml          (CI/CD - 145 lines)
├─ backend/
│   ├── Dockerfile            (44 lines - 3-stage build)
│   └─ .dockerignore
│
│   ├── prisma/schema.prisma  (257 lines - 12 models)
│   ├── package.json          (45 lines)
│   └─ src/
│
│       ├── app.js            (91 lines)
│       ├── index.js          (18 lines)
│       ├── modules/          (8 modules: auth, product, category,
│                               │   cart, order, hamper, address,
│                               │   user)
│       ├── config/           (db, mail, passport, redis)
│       ├── lib/webhook/       (Cashfree webhook handler)
│       └─ utils/
├─ frontend/
│   ├── Dockerfile            (44 lines - 3-stage build)
│   ├── .dockerignore
│   ├── next.config.ts        (with output: 'standalone')
│   ├── package.json
│   ├── app/                  (7 pages: home, shop, cart,
│                               │   checkout, auth, hamper, account)
│   ├── components/          (22 components)
│   ├── store/                (Redux - 7 slices)
│   ├── context/
│   ├── providers/
│   └─ types/
├─ k8s/
│   ├── namespace.yaml
│   ├── secrets.yaml
│   ├── backend-deployment.yaml (154 lines)
│   ├── frontend-deployment.yaml
│   └─ ingress.yaml

```



```

|   └─ hpa.yaml
└─ nginx/
|   └─ nginx.conf
└─ docker-compose.yml           (local dev)
└─ docker-compose.prod.yml      (EC2 production)
└─ DEPLOYMENT.md                (Step-by-step deployment guide)
└─ .gitignore

```

10.3 DevOps Pipeline Summary Table

Pipeline Stage	Tool	Duration	Trigger
Type Check	TypeScript + GitHub Actions	~1 min	git push main
Backend Image Build	Docker (3-stage)	~2 min	After lint passes
Frontend Image Build	Docker (3-stage)	~3 min	After lint passes
Push to ECR	AWS CLI + GitHub Actions	~1 min	After build
EC2 Deploy	SSH + docker compose	~1 min	After push, manual approval
Health Verify	curl /health	~30 sec	After deploy
Total Pipeline		~8–10 min	

CHAPTER 11 — BIBLIOGRAPHY

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End of Report

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